Preface from Author

From Static Diagrams to Executable Knowledge: A Personal Journey

For nearly thirty years, I have lived and breathed enterprise information technology. My career began long before “knowledge graph” was a common term, working across telecom, automotive, finance, and IT outsourcing consulting. For most of those years, my primary hands-on tools for making sense of complexity was static diagramming, mostly with Visio, draw.io, or even office suite like Excel and Powerpoint. It was powerful, but it was also… silent. The diagrams described architecture, but they could not execute it. They could not reason, validate, or adapt.

That was the before.

The shift (2021 – 2022) began when I moved from old toolboxes to Archi, the open-source ArchiMate modeling tool. Suddenly, I was not just drawing; I was building a **repository**. I discovered the power of a shared, collaborative model – a single source of truth about an enterprise. But even Archi, as good as it is, had limitations. The model was still primarily for human consumption. It was a rich, structured "graph" database, but it was not a reasoning engine.

It was around this same time that I discovered the **Essential EAS Open Source (Ontology)** – a glimpse of what happens when an EA repository is built on formal semantics. That was the first crack in the old paradigm, learnt and practiced with its modeling tool, Protégé - the first time I met this word.

The catalyst (2023) was the `Pizza.owl` tutorial. Through creating my first video series on the topic, I got in touch with Michael DeBellis. More importantly, I got my hands on Protégé. For the first time, I was not just modeling structure; I was formalizing **meaning**. I was defining classes, properties, and rules that a machine could use to infer new knowledge, check for contradictions, and navigate relationships intelligently.

The pizza example was simple, but the potential was explosive. I realized that the future of enterprise architecture was not in prettier diagrams, but in **executable knowledge**.

With that learning, back to the Essential EAS tool, I realized how powerful of treating enterprise architecture from semantic perspective, and how flexible the tool provided, e.g. if you need one

meaning (triple) that EAS built-in meta-model doesn't exist, just create one new from it's slots and that's now fitting the knowledge of your enterprise reality, more and more.

The synthesis & the birth of EKA & SKDL (2023 – 2026) became a period of intense exploration. My YouTube channel (and other video sharing platform, like BiliBili, DouYin, etc..) and online courses grew to host not just the `Pizza.owl` series, but courses on ArchiMate, meta-modeling, diagramming as code (PlantUML, D2, OpenSCAD), Neo4j graph databases, and ontology engineering. Each course was a piece of a larger puzzle. programming taught me logic, PlantUML taught me that diagrams could be generated from text (code), Neo4j showed me how to traverse relationships at scale, and OpenSCAD reinforced the power of parametric, declarative design.

Through years of distillation, everything crystallized into two simple, powerful core concepts for readers to take away:

1. **EKA (Executable Knowledge Architecture)**: The overarching paradigm that treats enterprise architecture and knowledge graphs not as static documents, but as executable, machine-readable specifications.

A slogan now circulating in industrial semantics is *Meaning before Automation*, associated with Frédéric Verhelst's *The Ontology Imperative*. It captures a readiness problem rather than a technology problem: agreed meaning is the bottleneck, and an ontology is that meaning in a form a machine can check. This book accepts the slogan and asks the engineering question it leaves open -- meaning must not only precede automation, it must be **executable, checkable and governable**. That is what EKA provides, and the `Pizza` ontology is the vehicle we use to show it.

2. **SKDL (Semantic Knowledge Development Lifecycle)**: The structured 7-stage approach I formulated to guide practitioners seamlessly from raw domain concepts all the way to advanced semantic reasoning and governance.

Why this book? This book is not a transcript of my videos. It is the theory behind the practice. It is the result of years of hands-on work, structured precisely around the SKDL stages. I wrote it because I saw a gap:

> *countless tutorials show you how to click buttons in Protégé, or write queries in Neo4j, but almost none provide a complete lifecycle methodology for architectural thinking.*

This book aims to bridge that gap, using the familiar `Pizza.owl` as a safe and repeatable domain to walk you step-by-step through the SKDL framework.

You will start with pizza. But if you follow the journey across the volumes, you will be able to model anything: an enterprise architecture, a digital twin, a regulatory framework, or an AI knowledge system.

The goal is not just to teach you a tool. The goal is to give you a new way to think about knowledge through **EKA** and **SKDL**: as something that can be **formalized, reasoned over, governed, and executed.**

Welcome to the journey.

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